Sai Krishna Singapati

+91 9949510434 • saikrishna96singapati@gmail.com • linkedin.com/in/sai-krishna-singapati •
saikrishna498.github.io

PROFESSIONAL SUMMARY

Built an Ethernet dual-bank OTA bootloader for STM32H5 at Q-PAC — completes updates in under 3 minutes, rollback-capable, and deployed across 1,000+ field devices. Over 5+ years I've shipped real-time motor control firmware (up to 9 EC motors), industrial protocol stacks (Modbus, BACnet), and CI/CD pipelines that catch memory bugs before they reach hardware. Experienced across the full lifecycle: board bring-up, DVT testing, layered firmware architecture, and Linux kernel-level driver debugging at Qualcomm. Most comfortable on ARM Cortex-M with FreeRTOS or Zephyr, close to the hardware.

TECHNICAL SKILLS

Languages	Proficient: Embedded C, C++ | Familiar: Python (automation & scripting)
Platforms	Proficient: ARM Cortex-M, STM32, ESP32 | Familiar: Cypress, DSPIC, Raspberry Pi
RTOS	FreeRTOS, Zephyr, CMSIS-RTOS
Protocols	CAN, SPI, I2C, UART, RS485, TCP/IP, Modbus, BACnet | BLE
Tools	JTAG, Logic Analyzers, Oscilloscopes, Wireshark
DevOps	Git, Jenkins CI/CD, CMake, GTest, Valgrind, GCOVR

WORK EXPERIENCE

Firmware Developer II
August 2023 – Present

Q-PAC Systems — Elkton, FL

- Architected and delivered an Ethernet-based **dual-bank OTA bootloader for STM32H5** — completes firmware updates in under 3 minutes with automatic rollback on failed verification; deployed and running reliably across **1,000+ field devices**.
- Integrated **Modbus and BACnet** (IP and serial variants) into a multi-protocol device stack, enabling interoperability across heterogeneous industrial environments.
- Developed **real-time EC motor control firmware** managing arrays of **up to 9 EC motors** simultaneously under varying loads — designed for fault tolerance with graceful degradation on sensor or comms failure.
- **Led board bring-up and DVT testing**; designed a layered firmware architecture that decoupled hardware abstraction from application logic, making it straightforward to add new sensor types and communication interfaces.
- Set up a **Jenkins CI/CD pipeline** with automated builds, GTest unit tests, static analysis, and GCOVR coverage reporting — caught several memory issues before they reached hardware.

System/Software Test Engineer · Embedded Focus
July 2021 – July 2023

Qualcomm — San Diego, CA

- **Debugged Linux kernel-level camera driver issues** on mobile chipsets — worked directly with firmware teams to isolate timing, race conditions, and memory problems in the camera subsystem.
- Validated camera subsystem drivers using Google's CTS/VTS/ITS conformance suites on ARM-based mobile SoCs.
- Wrote **Python automation scripts** that significantly reduced manual test cycle time and improved coverage consistency across regression suites.
- Contributed to design reviews and pushed for process improvements that reduced regression cycles.

Firmware Engineer
December 2018 – May 2019

DURA Automotive — Hyderabad, India

- Performed unit testing and validation of automotive gear shifter firmware using **LDRA**, targeting edge cases in state machine logic — code developed under **MISRA-C guidelines**; gained hands-on exposure to automotive SDLC, requirements traceability, and version control workflows.

PROJECTS

STM32H563 UART Configurable Console

saikrishna498.github.io · github.com/SaiKrishna498

Personal project — open source

- Built an **interrupt-driven UART console on STM32H563** featuring a TX queue, RX FIFO, command parser, and live baud-rate reconfiguration without halting the system — written in Embedded C using STM32 HAL and CMSIS.
- Demonstrates low-level peripheral control, ISR-safe data structures, and real-time reconfiguration patterns directly applicable to production firmware work.

EDUCATION

M.S. Electrical & Computer Engineering — Embedded Systems *August 2019 – May 2021*

San Diego State University San Diego, CA

GPA 3.67/4.0 · Relevant coursework: Embedded RTOS, Computer Networks, Cyber-Physical Systems

B.E. Electronics & Communication Engineering *August 2014 – May 2018*

Vellore Institute of Technology India

GPA 7.86/10 · Relevant coursework: C Programming, Data Structures, Microprocessors & Microcontrollers, Digital System Design, Computer Architecture, Computer Networks